

Fatra Al Khawarizmi

Jl. Pandanwangi, Cileunyi, Kab. Bandung • Bandung, 40625 • fatraalk@upi.edu • +6285724296661

Education

Universitas Pendidikan Indonesia

Candidate for Bachelor of Computer Science

Bandung, West Java
March, 2027

- GPA: 3.91
- Duolingo English Test: 120/160

Summary

Software Engineer student at Universitas Pendidikan Indonesia (GPA 3.91) with experience spanning software engineering, research, entrepreneurship, and international leadership programs. Recognized through international hackathons, research initiatives, and ASEAN youth development programs, with a strong interest in leveraging technology and cross-cultural collaboration to create social and economic impact.

Awards and Honors

- Top 6 Global & Top 1 Asia Dev3Pack International Hackathon
- 1st Place Outstanding Student Award, Faculty of UPI Cibiru Campus
- Fully Funded Delegate ASEAN Youth Conference Malaysia 2025
- Most Outstanding Scholar ASEAN Youth Organization Academy
- 4th Place National Winner Marketeers Innovation Challenge (MarkPlus & Pertamina)
- 1st Place UI/UX Design Competition, Itenas Digital Transformation Competition
- 3rd Place Incremental Business Plan Competition 2025

Research Experiences

- Co-authored a research manuscript, "*Implementation of IndoBERT SetFit for Automated Essay Scoring*", investigating few-shot learning with Back Translation and SMOTE augmentation for Indonesian-language essay assessment; submitted to a national informatics journal.
- Developed SAVIO, an AI-powered smart glove translating sign language into audio in real time; proposed for Indonesia's national Student Creativity Program (PKM-GFT).
- Co-developed VORCE, an AI-based voice and facial recognition system for mental health monitoring and intervention; proposed for Indonesia's national Student Creativity Program (PKM-GFT).

Professional Experiences

PT YAIJ Solusi Internasional

December, 2025 – Now

Fulltime Remote, IT Business Analyst

- Led end-to-end development of YAIJ: Cek Karir, generating **200+** qualified leads and attracting **3,000+** visitors during launch.
- Architected an enterprise management platform spanning **8+ operational modules**, **9 user roles**, and **4 external integrations**, streamlining workflows across multiple business functions.
- Partnered with the Project Manager to translate business requirements into scalable technical solutions, reducing data processing and content production time by **~60%** through automation, centralized data architecture, and workflow optimization.
- Established a containerized CI/CD infrastructure with GitHub Actions, Docker, and VPS deployment, supported by **90+ automated tests** to ensure application quality and deployment consistency.

Intern at Human Centric Research Center Telkom University

September, 2025 – January, 2026

Frontend Developer

- Developed dynamic conferencing applications with multi-role users, helping to simplify content creation with a 40% improvement
- Contributed to a research-based application registered as Intellectual Property (HKI), ensuring originality and legal protection of the developed system.

Universitas Pendidikan Indonesia

September, 2025 – January, 2026

Assistant Lecturer | MySQL (DDL, DML)

- Assisted lecturers in teaching Database Modeling, Basic Programming, and Software Design & Specification courses for 90+ students.

GABOOTHZ: Gallery Photobooth

February, 2025 – Now

Founder (Fullstack Developer, IT Business Analyst)

- Collaborated with 5+ stakeholders to deliver a seamless digital photobooth experience.
- Collaborated with 3 people to build Gaboothz and Generated more than IDR 10 million in revenue through commercial photobooth services.
- Selected as a finalist in UPI Edupreneur Academy's Business Entrepreneurship Program.

Organizational Experiences

ASEAN Youth Organization Academy

April, 2025 – Now

Strategic Partnership, Ex-Scholar

- The Most Outstanding Scholars of the AYO: ASEAN Youth Organization Academy
- Led 2 teams consisting of 8 members to develop a Business Model Canvas (BMC), business plan, and pitch deck for a mental health application.

Komunitas Satu Persen Bandung

August, 2024 – August 2025

Head of External Relations and Business Development

- Established partnerships with Bandung-based communities to co-host mental health and psychology awareness events, expanding organizational reach and cross-community engagement.
- Researched and mapped 141 schools across Bandung to drive the KSP Goes to School program, bridging high school students' interest toward higher education pathways.

Technical Projects

Fraud Detection System: SignalShield

February, 2026 – Now

AI Engineer | Business Analyst

- Developed and deployed a production-grade fraud detection platform used by a **payment gateway company**, leveraging Graph Analytics, Machine Learning, and Rule-Based Detection for real-time transaction monitoring and fraud prevention.
- Engineered **30+ rules (4-layer detection)**, achieving **95% money mule detection rate**
- Monitored **3,000+ entities** with real-time alerts & risk scoring
- Designed RBAC and Maker-Checker workflows aligned with OJK and Bank Indonesia governance requirements.
- Selected as **research contributor for Monash University** (Digital Financial Fraud Prevention Study) on fraud detection systems, regulatory challenges, and fintech security

JudgeChain

10 May 2026

Fullstack Developer

- Developed **100+ autonomous AI agents** with public, trader, and narrative enthusiast persona for decentralized decision-making and on-chain debate analysis on Solana, contributing to **a Top 6 Global** and **Top 1 Asia** finish at the Dev3Pack International Hackathon.

Top 40 Samsung Solve for Tomorrow

June, 2025 – August, 2025

Samsung Health Guard

- Developed an inclusive mobile health application promoting sports accessibility for women and persons with disabilities, aligned with SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities).

Technical Skills

- **Programming:** TypeScript, JavaScript, PHP, Python, SQL, Rust
- **Frameworks:** Next.js, React.js, Laravel, Express.js
- **Databases & Infrastructure:** PostgreSQL, MySQL, Docker, Nginx, GitHub Actions
- **AI & Data:** Machine Learning, Graph Analytics, Rule-Based Systems
- **Languages:** Indonesian (Native), English (Professional Working Proficiency)